



TAREA: 7.5

Hector.sH

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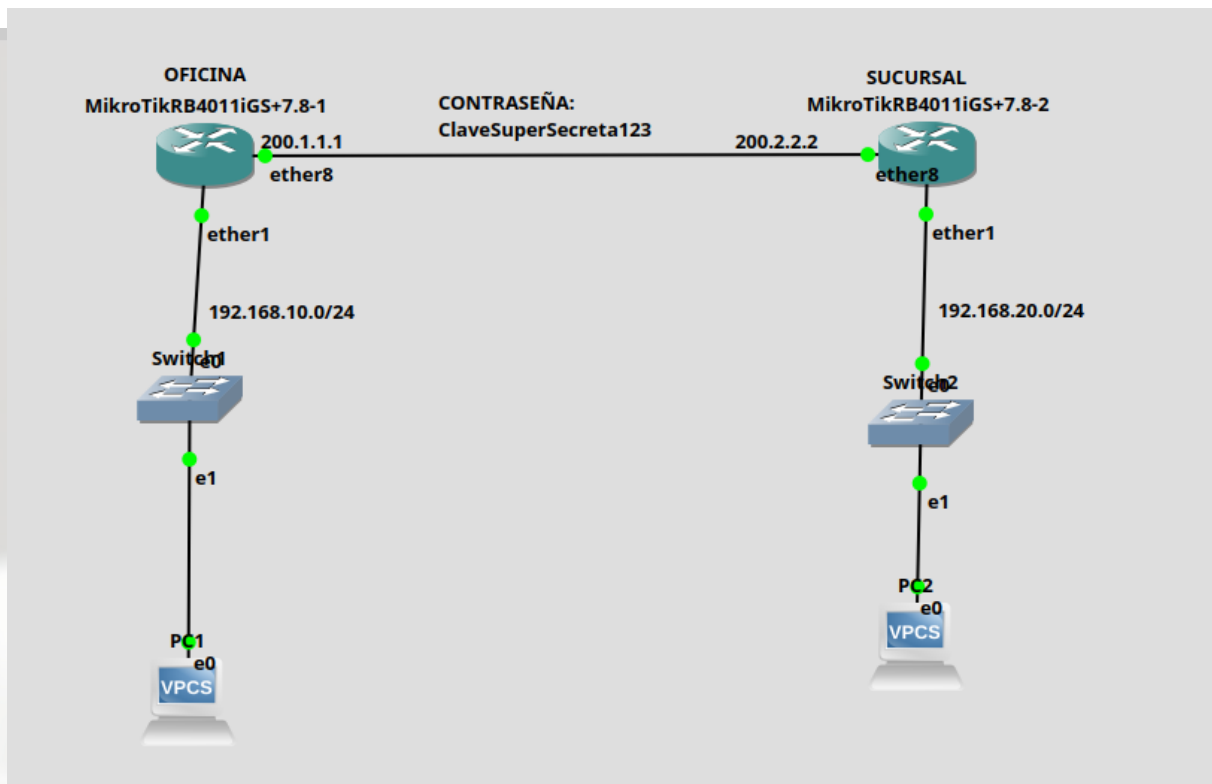


Figure 1: Esquema

```
[admin@MikroTik] > ip ipsec active-peers print
Flags: R - RESPONDER
Columns: STATE, UPTIME, PH2-TOTAL, REMOTE-ADDRESS
#  STATE      UPTIME  PH2-TOTAL  REMOTE-ADDRESS
0   established  35s           200.2.2.2
1 R established  39s           200.2.2.2
[admin@MikroTik] >
[admin@MikroTik] > ip ipsec installed-sa print
Flags: S - SEEN-TRAFFIC; E - ESP
Columns: SPI, STATE, SRC-ADDRESS, DST-ADDRESS, AUTH-ALGORITHM, ENC-ALGORITHM, ENC-KEY-SIZE
#  SPI      STATE  SRC-ADDRESS  DST-ADDRESS  AUTH-ALGORITHM  ENC-ALGORITHM  ENC-KEY-SIZE
0 SE 0x1B71A31 mature  200.2.2.2    200.1.1.1    sha256          aes-cbc        256
1 E 0x5B41303 mature  200.1.1.1    200.2.2.2    sha256          aes-cbc        256
2 E 0xA6BAA0A mature  200.2.2.2    200.1.1.1    sha256          aes-cbc        256
3 SE 0x98FFCFC mature  200.1.1.1    200.2.2.2    sha256          aes-cbc        256
[admin@MikroTik] >
```

Figure 2: Ipsec del router 1


```
[admin@MikroTik] > ip ipsec active-peers print
Flags: R - RESPONDER
Columns: STATE, UPTIME, PH2-TOTAL, REMOTE-ADDRESS
# STATE UPTIME PH2-TOTAL REMOTE-ADDRESS
0 established 36s 2 200.1.1.1
1 R established 31s 200.1.1.1
[admin@MikroTik] >
[admin@MikroTik] > ip ipsec installed-sa print
Flags: S - SEEN-TRAFFIC; E - ESP
Columns: SPI, STATE, SRC-ADDRESS, DST-ADDRESS, AUTH-ALGORITHM, ENC-ALGORITHM, ENC-KEY-SIZE
# SPI STATE SRC-ADDRESS DST-ADDRESS AUTH-ALGORITHM ENC-ALGORITHM ENC-KEY-SIZE
0 SE 0x98FFCFC mature 200.1.1.1 200.2.2.2 sha256 aes-cbc 256
1 E 0xA6BAA0A mature 200.2.2.2 200.1.1.1 sha256 aes-cbc 256
2 E 0x5B41303 mature 200.1.1.1 200.2.2.2 sha256 aes-cbc 256
3 SE 0x1B71A31 mature 200.2.2.2 200.1.1.1 sha256 aes-cbc 256
[admin@MikroTik] >
```

Figure 3: Ipsec del router 2

```
PC1> dhcp
DORA IP 192.168.10.199/24 GW 192.168.10.254

PC1> ping 192.168.20.199/24

84 bytes from 192.168.20.199 icmp_seq=1 ttl=62 time=26.708 ms
84 bytes from 192.168.20.199 icmp_seq=2 ttl=62 time=4.264 ms
84 bytes from 192.168.20.199 icmp_seq=3 ttl=62 time=3.411 ms
84 bytes from 192.168.20.199 icmp_seq=4 ttl=62 time=6.605 ms
84 bytes from 192.168.20.199 icmp_seq=5 ttl=62 time=3.434 ms
```

Figure 4: Ping desde el pc 1 al 2

1	0.000000	fe80::e83:b7ff:fe4d...	ff02::1	MNDP	210 5678 - 5678 Len=148
2	0.000001	200.1.1.1	255.255.255.255	MNDP	190 5678 - 5678 Len=148
3	0.000134	0c:83:b7:4d:00:07	CDP/VTP/DTP/PAgP/UD...	CDP	122 Device ID: MikroTik Port ID: ether8
4	0.000243	0c:83:b7:4d:00:07	LLDP Multicast	LLDP	151 MA/0c:83:b7:4d:00:00 IN/ether8 120 SysN=MikroTik SysD=MikroTik RouterOS 7.8 (stable) F
5	0.000806	fe80::e83:b7ff:fe4f...	ff02::1	MNDP	210 5678 - 5678 Len=148
6	0.000907	200.2.2.2	255.255.255.255	MNDP	190 5678 - 5678 Len=148
7	0.000962	0c:da:aa:4f:00:07	CDP/VTP/DTP/PAgP/UD...	CDP	122 Device ID: MikroTik Port ID: ether8
8	0.000984	0c:da:aa:4f:00:07	LLDP Multicast	LLDP	151 MA/0c:da:aa:4f:00:00 IN/ether8 120 SysN=MikroTik SysD=MikroTik RouterOS 7.8 (stable) F
9	11.789996	200.1.1.1	200.2.2.2	ISAKMP	134 Informational
10	11.794579	200.2.2.2	200.1.1.1	ISAKMP	134 Informational
11	11.807246	200.2.2.2	200.1.1.1	ISAKMP	134 Informational
12	11.811211	200.1.1.1	200.2.2.2	ISAKMP	134 Informational
13	16.835892	0c:83:b7:4d:00:07	0c:da:aa:4f:00:07	ARP	42 Who has 200.2.2.2? Tell 200.1.1.1
14	16.836784	0c:da:aa:4f:00:07	0c:83:b7:4d:00:07	ARP	42 200.2.2.2 is at 0c:da:aa:4f:00:07
15	16.866247	0c:da:aa:4f:00:07	0c:83:b7:4d:00:07	ARP	42 Who has 200.1.1.1? Tell 200.2.2.2
16	16.866720	0c:83:b7:4d:00:07	0c:da:aa:4f:00:07	ARP	42 200.1.1.1 is at 0c:83:b7:4d:00:07
17	59.998178	fe80::e83:b7ff:fe4d...	ff02::1	MNDP	210 5678 - 5678 Len=148
18	59.998235	200.1.1.1	255.255.255.255	MNDP	190 5678 - 5678 Len=148
19	59.998338	0c:83:b7:4d:00:07	CDP/VTP/DTP/PAgP/UD...	CDP	122 Device ID: MikroTik Port ID: ether8
20	59.998437	0c:83:b7:4d:00:07	LLDP Multicast	LLDP	151 MA/0c:83:b7:4d:00:00 IN/ether8 120 SysN=MikroTik SysD=MikroTik RouterOS 7.8 (stable) F
21	60.905706	fe80::e83:b7ff:fe4f...	ff02::1	MNDP	210 5678 - 5678 Len=148
22	60.906161	200.2.2.2	255.255.255.255	MNDP	190 5678 - 5678 Len=148
23	60.906410	0c:da:aa:4f:00:07	CDP/VTP/DTP/PAgP/UD...	CDP	122 Device ID: MikroTik Port ID: ether8
24	60.906697	0c:da:aa:4f:00:07	LLDP Multicast	LLDP	151 MA/0c:da:aa:4f:00:00 IN/ether8 120 SysN=MikroTik SysD=MikroTik RouterOS 7.8 (stable) F

Figure 5: Wireshark de la WAN

1	0.000000	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xed701197
2	4.172736	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xed701197
3	8.618146	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xed701197
4	10.780651	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xed701197
5	13.323398	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xed701197
6	14.826223	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xed701197
7	21.452789	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xed701197
8	26.401242	fe80::e83:b7ff:fe4d...	ff02::1	MNDP	210	5678 → 5678 Len=148
9	26.401488	192.168.10.254	255.255.255.255	MNDP	190	5678 → 5678 Len=148
10	26.401604	0c:83:b7:4d:00:00	CDP/VTP/DTP/PagP/UD...	CDP	122	Device ID: MikroTik Port ID: ether1

Figure 6: Wireshark de la LAN

La diferencia es que en la lan, practicamente estamos dominados por paquetes del dhcp, mientras que el la wan escuchamos muchos mas paquetes relacionados con la comunicacion entre los routers.

